

Rui Casaca

Software Engineer | Mission-Critical Systems & Geospatial Operations

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About me

Software Engineer with over 15 years of professional experience operating in high-stakes, mission-critical environments, combined with solid experience in modern software development (TypeScript, React, Java, Python). Proven track record of discipline and resilience, having completed a Computer Engineering degree at ISEC while serving in national emergency and geospatial operations (Portuguese GNR), followed by a Space Systems engineering internship at Critical Software. Specialized in translating complex operational requirements into reliable, data-driven software, real-time telemetry, and GIS/UAV solutions.

Technical Skills

Languages: TypeScript, Python, Java, C/C++, C#
Frontend & Mobile: React, HTML/CSS, Flutter, Kotlin, Angular
Backend: Node.js, ASP.NET Core (.NET 8), Entity Framework Core, REST API, WebSockets
Geospatial & Operations: ArcGIS Pro, QGIS, Spatial Analysis, Operational Planning, Field Drone Ops
Databases: Oracle SQL, PostgreSQL, SQLite, MongoDB, Firebase, Supabase
Tools: Git, GitHub, BitBucket, Docker, Postman, Confluence, AI-Assisted Engineering / Prompt Architecture
Cloud & Infrastructure: Azure Cloud, Google Cloud Run, Cloudflare
Architecture & Practices: JPA, Spring Boot, Microservices, Authentication, Agile, Software Validation, Requirements, ECSS standards

Professional Experience

Junior Software Engineer (Curricular Internship)

Critical Software — Feb 2026 - Jul 2026

Space Systems: Ground Segment, Telemetry, ECSS, CCSDS
Space Packets, Software Traceability, System Validation

- Built hands-on experience developing Lighthouse, a mission-critical Ground Segment platform for space operations.
- Engineered real-time telemetry visualization and data processing features, gaining practical experience integrating REST APIs and high-frequency WebSocket streams.
- Applied solid experience in strict software traceability, system validation, and CCSDS space packet structures under European space standards (ECSS).

Geospatial Solutions & Data Consultant (Freelance)

Fiverr, Upwork — Nov 2022 - Present

- Extensive experience designing and delivering custom GIS solutions, spatial data pipelines, and 3D terrain/asset models for international clients in operational contexts.
- Proven experience managing end-to-end project lifecycles, coordinating multidisciplinary teams (2–5 specialists), and translating complex operational requirements into reliable tools.

Project Manager, Developer & Operational GIS Analyst

GNR – UEPS (National Republican Guard – Emergency Protection & Relief Unit) — Dec 2010 - Mar 2019, Mar 2019 - Mar 2026

- Proven experience as Project Manager and Developer designing and deploying software solutions and GIS platforms to manage and coordinate multiple field teams at a national scale.
- Hands-on experience supporting missing persons search planning and disaster response operations through Geographic Information Systems (GIS) and field drone utilization, driving technological innovation in operational workflows.
- Over 15 years of professional experience in mission-critical environments requiring rapid decision-making, cross-functional teamwork, and strict operational accuracy.

Projects

Field Operations & Resource Optimization Platform

TypeScript, React, Node.js, Azure Cloud, Cloudflare, Supabase (PostgreSQL), OAuth/Biometrics

- Architected and deployed a production cloud platform for real-time field work tracking, task auditing, budget generation, and transport documentation with multi-role access control (Admin, Manager, Field Worker, Client).
- Engineered an automated workforce scheduling and routing engine using the Hungarian Algorithm, optimizing worker allocation across sites based on skill proficiency, geographical distance, and operational constraints.

Missing Persons Search & Operations Platform

React, TypeScript, Node.js, LLM APIs

- Applied real-world operational experience to develop a full-stack platform supporting missing persons search operations, enabling centralized information management and analysis.
- Integrated LLM APIs for contextual data analysis, implemented real-time dashboards, and built automated PDF report generation with email distribution.

HomeVault - Inventory Management App

Flutter, Riverpod, Drift (SQLite), Google Drive API

- Developed an Android asset and inventory management application with an offline-first architecture, demonstrating hands-on experience in secure local data persistence and cataloguing.
- Implemented local data management with Drift (SQLite), encrypted backups, Google Drive synchronization, biometric authentication, QR/barcode scanning, and Excel export.

Education

Bachelor's in Computer Engineering – Software Development Track

Instituto Superior de Eng. de Coimbra (ISEC) — 2023 - 2026

Completed while serving full-time in active national emergency operations.